ACTIVITY 1: DEVELOPMENT APPROACH AND REQUIREMENTS ELICITATION

Project: RP-Cloud Labs

Course: ITLSE801 - Software Engineering

1. Development Methodology Justification

The group selected the **Agile development methodology** for the RP-Cloud Labs Tool-Session Service. Agile is appropriate because the system's requirements may change as more colleges join the platform and request access to different learning tools. The methodology supports iterative development, continuous stakeholder feedback, collaboration, and regular testing.

The project will be divided into small development iterations. During the initial iteration, the team will focus on the core functions: submitting tool-session requests, validating student accounts, checking college license capacity, and recording approved sessions. Each iteration will include planning, implementation, testing, and review.

Agile also supports teamwork through GitHub, where members can work on separate branches, submit pull requests, review changes, and integrate approved code into the main branch. This approach helps identify defects early and improves the quality of the software.

Although Agile requires regular communication and active stakeholder participation, its flexibility and incremental delivery make it more suitable than Waterfall for this project. Therefore, the group recommends Agile to deliver a functional minimum viable product (MVP) within the available time and budget while allowing future improvements.

2. Stakeholder List

No.	Stakeholder	Interest and Responsibility
1	Students	Request access to hosted learning tools and use them through a web browser.
2	Lecturers	Use learning tools for teaching, practical exercises, and assessments.

3	College Administrators	Support college onboarding and manage college-related information.
4	RP Central IT Team	Maintain the platform, manage access, monitor performance, and ensure security.
5	Software and Tool Vendors	Provide software licensing information and license-capacity requirements.
6	Department Heads and Academic Managers	Ensure that the platform supports academic activities and institutional objectives.
7	Development Team	Analyze requirements, design, implement, test, and maintain the service.

3. Requirements Elicitation Techniques

The group selected interviews and questionnaires to collect requirements from the stakeholders.

3.1 Interviews

Target stakeholders: Lecturers.

Interviews allow the development team to ask direct questions and obtain detailed information about lecturers' teaching activities, practical assessments, software requirements, and challenges. Follow-up questions help clarify requirements and identify the features needed to provide a consistent learning environment.

3.2 Questionnaires

Target stakeholders: Students.

Questionnaires allow the team to collect feedback from many students across different colleges efficiently. They can identify students' preferred learning tools, difficulties with personal computers, internet-access challenges, and expectations regarding ease of use.

4. Five Notes Captured from the Mock Interview

Interview participant: Lecturer (role-play participant)

Purpose: To identify requirements for the Tool-Session Service.

1. Students use computers with different specifications, and some devices cannot run the required learning software properly.
2. Students should be able to access learning tools through a web browser without installing the software on their personal computers.
3. The system should provide a consistent software environment to support fair and reliable practical assessments.
4. The system should validate a student's account before granting access to a requested learning tool.

5. GitHub Repository and Collaboration

The group will use GitHub to store the project documentation and source code, coordinate team contributions, and review changes before integration.

Proposed repository name: **rp-cloud-labs-tool-session-service**

The repository README will describe the project overview, objectives, scope, technologies, and team collaboration process. Group members will work on separate feature branches and submit pull requests for review before their changes are merged into the main branch.

Branch protection will be configured for the main branch to require pull requests before merging. Where supported and required, the group will also enable review approvals and status checks.

GitHub repository URL: https://github.com/fredndabikunze01/rp-cloud-labs-tool-session-service.git